INCH-POUND

MIL-DTL-20268J
18 February 2025

SUPERSEDING
MIL-DTL-20268H(MC)
13 February 2017

DETAIL SPECIFICATION

FRAME, CAP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers service and dress cap frames.

1.2 Classification. The cap frames are of the following types, classes, and sizes, as specified (see 6.2):

1.2.1 Types and classes.

a. Type I – Black poromeric visor
   - (1) Class 1 – with black poromeric chin strap
   - (2) Class 2 – with gold chin strap

b. Type II – Black broadcloth visor
   - (1) Class 1 – with field grade embroidery and gold chin strap
   - (2) Class 2 – with field grade embroidery and black poromeric chin strap
   - (3) Class 3 – with general officer embroidery and gold chin strap
   - (4) Class 4 – with general officer embroidery and black poromeric chin strap

1.2.2 Sizes. The frame caps are of the following sizes. The recommended head measurements for each size are listed in 3.6.

Schedule of sizes

6	6-5/8	7-1/4
6-1/8	6-3/4	7-3/8
6-1/4	6-7/8	7-1/2
6-3/8	7	7-5/8
6-1/2	7-1/8	7-3/4

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AMSC N/A FSC 8405

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199	-	Thread, Polyester Core, Cotton or Polyester-Covered
A-A-52094	-	Thread, Cotton
A-A-59826	-	Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-B-3461	-	Button, Insignia, Metal, Uniform and Cap
MIL-E-17568	-	Embroidery Materials, Metallic and Synthetic Metallic
MIL-DTL-20271	-	Lace, Gold - Ornamental
MIL-DTL-29405	-	Cloth, Broadcloth: Wool and Wool/Nylon
MIL-DTL-32075	-	Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D256	-	Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D790	-	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1238	-	Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D1525	-	Standard Test Method for Vicat Softening Temperature of Plastics

ASTM D1593	-	Standard Specification for Nonrigid Vinyl Chloride Plastic Film and Sheeting
ASTM D2240	-	Standard Test Method for Rubber Property–Durometer Hardness
ASTM D2821	-	Standard Test Method for Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D4976	-	Standard Specification for Polyethylene Plastics Molding and Extrusion Materials
ASTM D6075	-	Standard Test Method for Cracking Resistance of Leather
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM E1282	-	Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design. The caps shall be the Marine Corps designs specified in figures 1 through 8, consisting of a circular inner headband, with crescent shaped, embroidered cloth or poromeric top visor centered at the front of the frame. The visors shall be properly molded with heat into the uniform crescent shape and shall be the same size for all sizes of frame. The set of the visor from the horizontal plane of the headband shall not be more than 35 degrees or less than 30 degrees. The visor shall be flanged outward and set so that the stitching does not touch the forehead. The lapped ends of the inner band shall be lapped 1±1/4 inch to the right or left of the center back, and fastened together with not less than two staples. The sweatband shall not be visible at the lower edge of the headband. The support, support holder, and adjustable steel wire grommet shall conform in all respects to the design, dimensions, and details specified in figures 2 and 3 and as specified herein. The crown support stay holder shall be vertically straight, flush, and secure against the headband. Each cap frame shall have a chin strap attached to the cap with two insignia buttons inserted through the lower sides of the headband into their respective screw tubes and securely attached. A hole shall be punched through the left and right sides of the headband to allow for the button screw tubes. The holes shall be positioned so that the center of the screw tube on both sides of the finished frame shall be 5-3/4±1/16 inches from center of visor and 5/8±1/16 inch from the lower edge of headband. Each frame shall be supplied with a clear acetate crown protector (see 3.4.12).

3.4 Materials.

3.4.1 Inner band. The inner band shall be of a high density (linear) polyethylene plastic material, in black shade, conforming to group 2, high density class 3 or 4, grades 1 or 2, of ASTM D4976, except that the low temperature brittleness, dielectric constant, dissipation factor, and thermal stress cracking resistance requirements shall not be applicable. The band material shall be extruded, stripped, slit, or cut 2-1/4±1/16 inches in width and 0.030±0.002 inch in thickness. Testing shall be as specified in 4.4.

3.4.1.1 Stiffness. The inner band material shall have a stiffness (scale reading) between 11 and 27 inclusive when tested as specified in 4.4.1.

3.4.1.2 Bending. The inner band material shall show no evidence of cracks after the bending test specified in 4.4.2.

3.4.2 Headband outer covering. The headband outer covering shall be made from a commercially available gabardine, 100 percent polyester, 6.8 ounce per square yard, shade black. As an alternate, 6.0-ounce wool-synthetic broadcloth in a black shade may be used. The cloth shall be cut 3-3/4±1/8 inches wide in the warp or filling direction and cut to the proper length to snugly fit around the headband.

3.4.3 Sweatband. The sweatband shall be made from artificial leather composed of a nonwoven synthetic substrate impregnated with an elastomeric binder. The top surface of the substrate shall be coated with an abrasive resistant finish. The sweatband shall finish 1-7/8±1/16 inches in width. The sweatband shall be dyed black.

3.4.3.1 Sweatband seam reinforcement. The seam joining the ends of the sweatband shall be reinforced with a 5/8- to 3/4-inch wide strip of black pyroxylin coated cotton fabric cut the width of the sweatband, or the plastic film cited in 3.4.3.2 may be used.

3.4.3.2 Plastic film. The plastic film for the sweatband shall be 0.010 inch (nominal) in thickness, conforming to type I of ASTM D1593. The plastic film shall be cut 5/8 to 3/4 inch wide and the full length of the sweatband; the color shall be black.

3.4.4 Visor. The visor shall be of one size only. The overall width of the visor, before assembly, at center, shall be 2-5/8±1/8 inches. The depth of arc, at center of back edge, shall be 2-7/8±1/8 inches. The width of the visor, after assembly, at center from front edge to lower edge of band shall be 2-1/4±1/8 inches. The distance from the left tip to right tip of visor before assembly, when spread flat shall be 8±1/8 inches. The distance from left tip to right tip after assembly, at lower edge of headband, shall be 11±1/4 inches. The visor shall be constructed from the materials specified in 3.4.4.1 through 3.4.4.2.

3.4.4.1 Type I. The type I visors shall consist of a top ply, bottom ply, and reinforcement plumper paper, bonded together, with binding over the edge (see figure 1, type I). As an alternate to the plumper paper, polyethylene interlining with a gauge of approximately 0.050 inch may be used. The visor shall be constructed by laminating the top ply of poromeric material to the plumper paper and the bottom leather ply with the adhesive specified in 3.4.5.

3.4.4.1.1 Top ply. The top ply shall be smooth, high-gloss poromeric material 0.045±0.003 inch thick, 16-1/2±1 ounces per square yard.

3.4.4.1.2 Bottom ply. The bottom ply shall be of reconstituted leather in a black pebbled gloss finish and shall be 6 to 6-1/2 ounces thick. As an alternate, artificial leather coated with an abrasive resistant finish with a nominal thickness of 0.040 inch may be used, or 0.050 gauge polyethylene conforming to the properties in table I.

TABLE I. Polyethylene properties.

Properties	Reference paragraph	Requirement
Density, grams per cubic centimeter (g/cm^3)	4.3.1	0.944 – 0.976
Melt index, I-2, grams in 10 minutes	4.3.1	0.36 – 0.44
Flexural modulus, pounds per square inch (psi)	4.3.1	153,000 – 187,000
Tensile yield strength, psi	4.3.1	4,050 – 4,950
IZOD impact strength (notched), foot-pounds of force per foot (ft·lbf/ft)	4.3.1	1.08 – 1.32
Vicat softening point, °F	4.3.1	249 – 275
Hardness Shore D	4.3.1	67 – 81

3.4.4.1.3 Plumper paper. The plumper paper for laminating to the top and bottom plies of the visor shall be 20 gauge and treated so that it retains its original shape after being subjected to immersion test. Plumper paper is not required if the bottom ply is constructed from the 0.050 gauge polyethylene listed as an alternate in 3.4.4.1.2.

3.4.4.2 Type II. The type II visors shall consist of a top ply of black broadcloth conforming to type I of MIL-DTL-29405 and a bottom ply of reconstituted leather or artificial leather in hatters green or black pin seal gloss finish. The visor shall be constructed by laminating the top ply of wool broadcloth to the reinforcement piece of cloth cotton twill face side up with the adhesive specified in 3.4.5. The top ply shall be embroidered with tarnish resistant synthetic gold yarn conforming to type III, class 8 of MIL-E-17568. The design of the embroidery shall be as shown in figure 1 for the appropriate class.

3.4.5 Adhesive. The adhesive shall be used to cement the component parts of the visor together. The adhesive shall be either natural or synthetic rubber cement. Lamination strength shall be a minimum of 5 pounds per inch.

3.4.6 Binding. The binding for the visor edge shall be made from a plastic film, weighing not less than 12 ounces per square yard, and having a thickness of not less than 0.0013 inch. The binding shall be cut 3/4 to 7/8 inch wide in the appropriate length to fully bind the outside edge of the visor. The binding shall be stitched around the outside edge of both type I and II visors and shall finish on top of the visor 3/16 to 1/4 inch wide. The plastic film shall have a smooth high finish, and the color shall match that of the top piece for the visor. The binding shall not crack and shall show no evidence of tackiness. The binding shall be tested as specified in 4.3.1.

3.4.7 Support stay and holder. The support stay and holder shall be stamped from straight chromium stainless steel, commercially designated as type 430 stainless steel with material composition as shown below when tested as specified in 4.3.1. The support, and support holder, shall conform in all respects to the design, dimensions, and details shown in figure 2 and as specified herein.

Composition:	Carbon	0.12% maximum (ladle analysis [L.A.])
	Manganese	1.00% maximum (L.A.)
	Silicon	1.00% maximum (L.A.)
	Chromium	14.00% to 18.00% (L.A.)
	Phosphorus	0.040% maximum (L.A.)
	Sulphur	0.030% maximum (L.A.)
	(Remainder iron and constituents)	

3.4.8 Buttons. The screwpost button shall be 27-ligne conforming to MIL-B-3461 and as specified in table II. The button screw tube shall be 1/4 inch in length.

TABLE II. Buttons.

Cap	Button
Type I, classes 1 and 2	Type II, style 3, class A, subclass 1, gold
Type II, classes 1 and 3	
Type II, classes 2 and 4	Type II, style 3, class B, subclass 1, black

3.4.8.1 Washers. The tinnerman lock washers for securing the screw tube to the frame shall be of spring steel with internal locking flanges to grip the screw tube. As an alternate, heavy duty wire staples may be used to secure the screw tubes.

3.4.9 Grommet. The grommet wire shall be high grade spring steel, 5/16 inch wide, 0.020±0.002 inch thick, with a white pyroxylin coating not less than 0.003 inch thick. See figure 3 for assembled view. The grommet wire shall be cut as follows:

30-1/4 inches long for sizes 6, 6-1/8, and 6-1/4;

32-1/4 inches long for sizes 6-3/8, 6-1/2, 6-5/8, and 6-3/4;

33-3/4 inches long for sizes 6-7/8, 7, 7-1/8, and 7-1/4; and

35-1/2 inches long for sizes 7-3/8, 7-1/2, 7-5/8, and 7-3/4.

3.4.9.1 Clamp. The adjustment clamp clinched to one end of the steel grommet wire shall be made of the following materials:

Base and clamp	-	Carbon steel American Iron and Steel Institute (AISI) 1020, hardness Rockwell B scale 66, thickness 0.031±0.002 inch.
Assembled clamp	-	Electroplated with zinc 0.00005-inch minimum thickness, chrome plated finish.

The clamp assembly shall be approximately 1 inch long and its design shall conform to figure 4. The width of the base component shall be sufficient to accommodate the width of the grommet wire. Testing shall be performed as specified in 4.3.1. Alternate clamps of the same or greater holding power of the grommet wire may be used.

3.4.10 Rivets. The rivets for securing the grommet to the support stay shall be nickel brass, with a head measuring 3/32 inch (minimum) in diameter, when tested as specified in 4.3.1. The length of the rivets shall be sufficient to provide a roll over edge of 0.045 inch to 0.050 inch. Steel rivets of alternative sizes and lengths may be used to attach the grommet wire providing the head of the rivet does not exceed 0.040 inch and the rolled-over edge presents a smooth clinch with no more than three splits in the coil.

3.4.11 Staples. The staples shall be stamped from a heavy-duty corrosion-resistant flat steel wire 0.0250±0.005 inch in thickness, and shall be of sufficient length to thoroughly secure the applicable component parts and leave a smooth flat clinch without any distortion of the materials. The staples shall have a corrosion-resistant finish that shows no evidence of corrosion during the life of the item, or shall be made of a stainless steel.

3.4.12 Crown protector. Each cap frame shall be furnished with a crown protector for the cap cover. The crown protector shall be made of clear, colorless acetate sheeting 0.008±0.001 inch thick and shall be oval shaped, measuring 10 inches by 9-1/2±1/8 inches.

3.4.13 Thread.

3.4.13.1 Thread, cotton. The cotton thread for seaming and stitching the cap frame shall conform to type III, 3-ply, ticket No. A; type III, 3-ply, ticket Nos. A and 00; and type II, 4-ply, ticket No. 24, of A-A-52094. The color of the thread shall be vat dyed black AA. The dyed thread shall show good colorfastness to laundering, perspiration, wet-dry cleaning, and light. Thread, polyester, cotton-covered, size 50, conforming to A-A-50199 may be used.

3.4.13.2 Thread, nylon. The nylon thread for seaming and stitching the cap frame shall conform to type II, size 3, 3-ply, of A-A-59826. The color of the thread shall be vat dyed black AA. The dyed thread shall show good colorfastness to wet-dry cleaning and perspiration.

3.4.13.3 Tarnish-resistant. The thread for embroidering the visors on the type II caps shall be tarnish resistant synthetic gold yarn conforming to type III, class 8 of MIL-E-17568.

3.4.14 Chin straps.

3.4.14.1 Black chin straps. The black chin straps for the type I, class 1 and type II, classes 2 and 4 caps shall be made of a single ply of poromeric material specified in 3.4.4.1.1. Chin straps shall be made in two sections with a hole at one end and a loop at the other end of each section. Ends of the loop shall be lapped and fastened with two staples through the end of the loops only. A centered, 3/16-inch diameter hole shall be punched into the strap, 3/8 inch from the rounded end of each strap. The top and bottom edges of the strap and the point of the loop shall have an embossed line. All edges shall be stained black. The chin strap design shall be in accordance with the requirements specified in figure 5 which specifies dimensions and construction requirements.

3.4.14.2 Gold chin straps. The gold chin straps for the type I, class 2 and type II, classes 1 and 3 shall be gold laced conforming to class I of MIL-DTL-20271, 5/8 inch wide, constructed with tarnish resistant synthetic yarn conforming to type III, class 8 of MIL-E-17568. The gold lace shall be mounted on leather or other leather-like material to provide body to the strap. Chin straps shall be made in two sections with an eyelet at one end and a loop at the other end of each section. A 3/16-inch inside diameter eyelet shall be inserted through the ends of each strap, opposite the loop end, with the center of the eyelet finishing 3/8 inch from the pointed end. The chin strap design shall be in accordance with the requirements specified in figure 5 which specifies dimensions and construction requirements.

3.4.15 Labels. Each cap shall have a cloth combination identification-size label and a label tag, conforming to 3.4.15.1 and 3.4.15.2.

3.4.15.1 Identification-size label. The combination identification-size label shall conform to type VI, class 4 of MIL-DTL-32075. The label shall bear the following inscription:

NOMENCLATURE: FRAME, CAP
CONTRACT NO.: SPE1C1-00-0-0000 (Example)
STOCK NO.: 8405-00-000-0000 (Example)
SIZE: 7 (Example)
NAME OF CONTRACTOR

3.4.15.2 Label tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper used for the tags shall conform to type VIII, class 17 of MIL-DTL-32075. The tags shall be located so that they are completely visible on the item when it is packaged as specified (see 5.1). The label's location and attachment shall cause no damage to the item.

3.5 Patterns. When specified (see 6.2), the patterns for cutting all parts of the cap frame shall be furnished by the contractor and shall be of the proper proportions to provide good fitting cap frames conforming to the finished dimensions in table III. The patterns shall conform to the constructional requirements specified in 3.7 and to the design requirements specified in 3.3.

3.6 Dimensions. The finished frame shall conform to the size and head measurements specified in table III. The head measurement shall be the inside circumference of the headband at the lower edge of the sweatband. The head measurement of each size frame shall be as specified in table III.

TABLE III. Head measurements.

Size of frame	Head measurements (circumference) [1/]	
	(inches)	(cm)
6	19	48
6-1/8	19-3/8	49
6-1/4	19-3/4	50
6-3/8	20-1/8	51
6-1/2	20-1/2	52
6-5/8	20-7/8	53
6-3/4	21-1/4	54
6-7/8	21-5/8	55
7	22	56
7-1/8	22-3/8	57
7-1/4	22-3/4	58
7-3/8	23-1/8	59
7-1/2	23-1/2	60
7-5/8	23-7/8	61
7-3/4	24-1/4	62

NOTE:
[1/] All head measurements ±1/8 inch (3.2 mm).

3.7 Construction.

3.7.1 Manufacturing requirements. The frames shall be manufactured in accordance with the operations and the stitch, seam, and stitching types specified in table IV.

3.7.1.1 Stitches, seams, and stitching. All stitching shall conform to ASTM D6193. The frames shall be constructed in accordance with the stitch types and stitches per inch specified in table IV.

TABLE IV. Stitching.

Operation	Stitch type	Seam/stitch type	Stitches per inch
Topstitch, single needle stitch, or straight stitch:	301		
Construct visor		BSb-1	6-9
Construct poromeric chin strap		SSa-2	8-10
Construct gold chin strap		SSa-1	6-10
Construct headband outer covering:			
Fold lengthwise and stitch 1/16 inch from fold forming welt 2±1/16 inch from top edge of finished cover		Osf-1	10-14
Join ends with 3/8-inch seam		SSa-1	10-14
Attach headband outer covering to inner headband		SSa-1	6-10
Attach sweatband to band		EFd-1	6-10
Attach visor to frame		LSa-1	4-8
Overedge:			
Serge top and bottom edge of headband outer cover	503 or 504	Efd-1	8-12
Zig Zag:	304		
Construct sweatband assembly:			
Stitch plastic welt strip (folded in half lengthwise) to sweatband; edge of sweatband shall extend 1/16 inch to 1/8 inch from turned edge of strip.		FSb-1	10-14
Join ends and attach reinforcement		FSe-1	10-14
Form loop around gold chin strap and attach to create loop	Hand		4-6
Visor embroidery (as applicable)	Hand		

3.7.1.2 Tacking and backstitching. Ends of seams and rows of stitching with 301 type stitch, when not caught in other seams or stitching, shall be securely backstitched not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches and thread breaks may be repaired using stitch type 301 backstitched not less than 1/2 inch on each side of skip or break.

3.7.1.3 Overedge stitching. Overedge stitching shall be done on serging machines with the knife attachment(s) properly set to trim the raveled edge without cutting the material.

3.8 Workmanship. The finished frame shall conform to the quality and grade of product established by this specification. It shall be free from any irregular defects that can adversely affect usage or durability, defects that adversely affect form, fit, or function, and those defects specified in 4.3.2.1.

3.9 Figures. Figures 1 through 6 specify requirements for the frame and the frame's components. Figures 7 and 8 are furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall govern.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the cap frame when a first article sample is required (see 3.1). This inspection shall include conformance to the dimensions specified in 3.6, the examination of defects specified in 4.3.2.1, and the tests in 4.4.1 and 4.4.2.

4.3 Conformance inspection. Conformance inspection shall include conformance to the dimensions specified in 3.6, the examination of defects specified in 4.3.2.1, and the tests of 4.4.1 and 4.4.2. Unless otherwise specified (see 6.2), sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4.

4.3.1 Testing of components. Components and materials in table V shall be tested in accordance with all the requirements of referenced specifications, figures, and standards unless otherwise excluded, amended, or modified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, figures, and standards, testing shall be performed on components listed in table VI for characteristics noted. Test reports shall be expressed as specified (see 6.2).

TABLE V. Component materials.

Component	Characteristic	Requirement paragraph
Inner headband	Material identification	3.4.1
Headband outer cover and sweatband welt	Material identification	3.4.2
Sweatband	Material identification	3.4.3
Sweatband seam reinforcement	Material identification	3.4.3.1
Visor, top piece type I	Material identification	3.4.4.1.1
Visor, bottom piece	Material identification	3.4.4.1.2
Visor, top piece type II	Material identification	3.4.4.2
Plumper paper or polyethylene interlining	Material identification	3.4.4.1.3
Adhesive	Material identification	3.4.5
Binding	Material identification	3.4.6
Rivets	Material identification	3.4.10
Staples	Material identification	3.4.11
Chin strap	Material identification	3.4.14

TABLE VI. Test methods.

Component	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as
Inner band	Thickness	3.4.1	Micrometer	2	To nearest 0.001 inch
	Width	3.4.1.1	Gauge	1	Pass or fail
	Stiffness	3.4.1.1	4.4.1	1	Pass or fail
	Bending	3.4.1.2	4.4.2	1	Pass or fail
Binding	Weight	3.4.6	Scale	2	To nearest ounce/yard
	Thickness	3.4.6	Gauge	2	To nearest 0.001 inch
	Finish	3.4.6	Visual	1	Pass or fail
	Color	3.4.6	Visual	1	Pass or fail
Staples	Thickness	3.4.11	Gauge	2	To nearest 0.001 inch
Grommet wire	Material identification	3.4.9	Standard Commercial	X	Pass or fail
	Thickness of wire	3.4.9	Standard Commercial	X	Average of 2 determinations per sample unit to nearest 0.001 inch
	Thickness of pyroxylin	3.4.9	Standard Commercial	X	Average of 3 determinations per sample unit to nearest 0.001 inch
Spring steel	Material identification	3.4.9	Standard Commercial	-	Pass or fail
	Temper	3.4.9	Standard Commercial	-	Pass or fail
Support stay and holder (50 pounds)	Type 430 stainless steel	3.4.7	ASTM E1282 2 Composite	X	To nearest 0.01% of each element
Rivets	Material identification and plating	3.4.10	Standard Commercial	X	Pass or fail
Bottom ply - polyethylene	Density	3.4.4.1.2	ASTM D792	X	Grams/centimeters^3
	Melt index, I-2	3.4.4.1.2	ASTM D1238, 190/2.16	X	Grams/10 minutes
	Flexural modulus	3.4.4.1.2	ASTM D790, I/B	X	Pounds/inch^2
	Tensile yield strength	3.4.4.1.2	ASTM D638	X	Pounds/inch^2
	IZOD impact strength (notched)	3.4.4.1.2	ASTM D256	X	Feet-pound-force/inch
	Vicat softening point	3.4.4.1.2	ASTM D1525	X	Degrees Fahrenheit
	Hardness shore D	3.4.4.1.2	ASTM D2240	X	Pass or fail

4.3.1.1 Sampling for testing. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with table VII. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified herein.

TABLE VII. Material sampling.

Lot size (lot unit)	Sample size (sample unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

The unit for expressing lot size and sample unit for testing each component shall be in accordance with the applicable subsidiary specification and table VIII as follows:

TABLE VIII. Component sampling.

Component	Lot size expressed as	Sample unit for testing
Inner band	feet	1 yard strip
Sweatband seam reinforcement	yard	1 foot length
Binding	10 yards	1 square foot
Rivets	gross	3 each
Staples	gross	10 each
Grommet	10 complete units	1 unit
Clamp	gross	10 each
Crown support and holder	100 complete units	1 complete unit
Button, button screw tube	gross	10 complete units

4.3.2 Examination of end item. The defects found during examination of the end item shall be classified in accordance with 4.3.2.1. The sample unit shall be one cap frame.

4.3.2.1 Defects. Defects shall be classified in accordance with table IX.

TABLE IX. Defects.

Examine	Defect	Classification	
		Major	Minor
Material defects and workmanship [1/]	Headband covering, sweatband welt:		
	Any hole or weakening defect, such as smash, multiple float, or loose slub that may develop into a hole.	101	
	Tears, mend, burn, cut, and needle chew.	102	
	Shade bar or unsightly slub:		
	On outside.		201
	On inside.		202
Cleanness	Any spot or stain clearly noticeable:		
	On outside.	103	
	On inside.		203
	Five or more thread ends not trimmed, or loose thread ends not removed.		204
Components and assembly	Any component or required operation omitted.	104	
	Any component not as specified.	105	
Cutting	Any component part not cut in accordance with specification requirements.	106	
Seams and stitching	Accuracy of seaming:		
	Seams twisted, puckered, or pleated, affecting appearance.		205
	End of stitching (when not caught in other seam or stitching) backstitched less than 1/4 inch.		206
	Stitching of thread breaks overlapping less than 1/2 inch at each end of break.		207
	Any component part caught in an unrelated operation or stitching.	107	
	Wrong color thread.		208
	Gauge of stitching:		
	Unevenly gauged, affecting appearance (to be scored only when the condition exists on major portion of seam):		
	On visor.	108	
	On all other components.		209

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Seams and stitching (continued)	Edge or raised stitching sewn too close to edge, resulting in damage to component.	109	
	Beyond range of width specified or varies more than 1/16 inch when no range is specified.		210
	Open seams 2/:		
	Any seam open on visor joining seam, visor binding, sweatband joining to frame, back seam of sweatband, or back seam of cloth band.		
	Less than 1/4 inch.		211
	More than 1/4 inch.	110	
	On all other seams:		
	Less than 1/4 inch.		212
	1/4 inch up to 1/2 inch inclusive.		213
	More than 1/2 inch.	111	
	Raw edges 3/:		
	Any raw edge on binding on upper edge of visor or on cloth at bottom edge of cap frame.	112	
	Run-offs:		
	On joining seam – when resulting in an open seam, use "Open seam" classification.		214
	Edge or raised stitching (when not resulting in an open seam):		
	1/4 inch up to 1/2 inch inclusive.		215
	More than 1/2 inch.		216
	Seam and stitch types:		
	Not specified seam or stitch type.	113	
	Stitch tension:		
	Loose tension resulting in a loose seam or exposed loose top or bottom thread:		
	On visor joining seam or on stitching of visor binding.	114	
	On all other seams or stitching.		217
	Tight tension (stitches break when normal strain is applied to the lengthwise direction of seam or stitching). 4/	115	

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Seams and stitching (continued)	Stitches per inch (to be scored only when the condition exists on major portion of seam).		
	Less than minimum specified except on hand felling of sweatband:		
	1 stitch.		218
	2 stitches.		219
	3 or more stitches.	116	
	Less than minimum specified on hand felling sweatband:		
	1 stitch.		220
	2 stitches.		221
	3 or more stitches.	117	
Marking	Combined identification and size label missing, incorrect, or illegible.		222
Visor	Not securely or evenly stitched with sweatband welt to inner headband.	118	
	Set from vertical front of cap at an angle of more than 35 degrees or less than 30 degrees.		223
	Center of visor offset by more than:		
	3/8 inch.	119	
	1/4 to 3/8 inch.		224
	1/8 to 1/4 inch.		225
	Skiving omitted on inside edge.		226
	Front edge uneven or not molded to proper shape.	120	
	Binding not securely caught to edge of visor.	121	
	Binding not firmly and smoothly joined to visor edge, affecting appearance.	122	
	Binding is peeling, cracked, tacky, spotted, discolored, or blistered.	123	
	Finished width of binding on top of visor under 3/16 inch or over 1/4 inch (but of uniform width).	124	
	Binding not of uniform width on top of visor, affecting appearance.	125	
	Width of visor at center less than 2-3/16 inches or more than 2-5/16 inches.		227
	Distance between ends of attached visor at headband less than 10-3/4 inches or more than 11-1/4 inches.	126	
	Poorly laminated, affecting appearance.	127	
	Visor joining seam, more than 3/16 inch or less than 1/16 inch from lower skived edge.		228

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Embroidery (type II)	Omitted.	128	
	Not specified type.	129	
	Core yarn not fully covered with metallic ribbon.		229
	Loose stitches.		230
Chin straps and loops	Length of each section under 10 inches or over 10-1/2 inches.		231
	Width of each section under 9/16 inch or over 11/16 inch.		232
	Edges of straps or points or loop not embossed. (type I, class 1 or type II, class 3). Embossing acceptable if visible at distance of 18-24 inches.	130	
	Edges not stained or poorly stained.		233
	Loop not securely stitched to top strap or not stapled with two staples.	131	
	Bottom section of loop not securely stapled to bottom strap or not stapled with two staples.	132	
	Length of loops under 1-1/8 inch or over 1-3/8 inches.		234
	Center of hole under 11/32 inch over 13/32 inch from end of strap.		235
	Size of eyelet opening or punched hole under 5/32 inch or over 1/4 inch.		236
	Holes off center of strap over 1/32 inch.		237
	Opening of loop narrow, curling edge of strap.	133	
Buttons and screw tubes	Center of tube under 9/16 inch or over 11/16 inch from lower edge of headband.		238
	Distance of screw tubes from center of crown support less than 5-5/8 inches or more than 5-7/8 inches.	134	
	Any button or screw tube omitted.	135	
	Tube(s) not securely locked.		239
	Button or screw tubes defective, affecting function.	136	
Headband, outer covering	Welt omitted.	137	
	Back seam not spread open or partially lapped over.		240
	Back seam offset from center back of frame by more than 1/4 inch.		241
	Back seam not in vertical position by more than 1/8 inch.		242
	Outer band badly puckered, pleated, or twisted, affecting appearance.		243

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Headband, outer covering (continued)	Raised, irregular welt.		244
	Outer band irregular in width by 1/8 inch or over (to be scored only when condition exists on major portion of band).		245
Inner band	Crushed, broken, warped or misshapen, affecting serviceability or appearance.	138	
	Overlapped ends insecurely fastened.		246
	Overlap not positioned either to the right or left of the center back.		247
	Width under 2-3/16 inches or over 2-5/16 inches.		248
	Ends overlapped under 3/4 inch or over 1-1/4 inches.		249
Grommet (with clamp)	Not fabricated of pyroxylin coated spring steel.	139	
	When wire depressed together into figure 8 position, and released, it does not spring back to its original circular shape.	140	
	Coating is not smooth and adherent. Coating contains beads, blisters, lumps, specks, pits, or foreign matter.	141	
	Coating is missing, scratched, scuffed, abraded, or otherwise damaged, in area of:		
	1/16 inch or more in width or 1 inch or more in length.	142	
	More than 1/32 inch, but less than 1/16 inch in width or more than 1/2 inch, but less than 1 inch in length.		250
	1/32 inch or less in width or 1/2 inch or less in length.		251
	Not clean.		252
	Length 1/2 inch more or less than required for size of cap.	143	
	Clamp hinge not on inside of grommet wire.	144	
	Clamp not securely clinched to wire.		253
	Clamp does not function properly.		254
	Rough or sharp edges on wire or clamp.		255

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Support stay and holder:			
Finish	Dirty. Surface contains grease, mill soil, or other foreign matter.	145	
Design	Any operation not in accordance with specified requirements.		256
Construction and workmanship	Component missing, cracked, fractured, bent out of shape, distorted, malformed, deformed, or otherwise impaired.		257
Assembly	Not connected or joined as specified or poorly accomplished.		258
	Sharp edge, burr or sliver that may injure personnel or damage fabric.	146	
	Rivet missing, wrong type or size, or is not peened. Rivet can be removed by hand.	147	
	Rivet head broken off.	148	
	Clinch of rivet incomplete, insufficient, or set improperly. Rivet is loose/can be rotated by hand or tool.	149	
	Flexing of crown wire causes rivet to loosen or wire to separate.	150	
	Rivet insufficiently peened but it cannot be removed by hand.		259
	Setting of rivet is poorly accomplished, resulting in sharp, rough edges.		260
	Rivets are not placed equidistant from edges of the stay ±1/32 inch.		261
	Rivets are not 5/8±1/32 inch apart, center to center.		262
	Stay and support holder will not mate, or a loose fit is affected.		263
	Assembled height less than 4 inches or more than 4-1/4 inches.		264

TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Sweatband	Edges of back seam out of alignment with outer covering seam by 1/4 inch or more.	151	
	Strip on underside of back seam omitted or not securely stitched.		265
	Strip not trimmed even with top edge of sweatband.		266
	Poorly felled to outer band and welt. Stitching of sweatband on outerband and welt is very irregular in appearance.		267
	Ends not abutted properly.		268
	Felling stitches exposed on outside of frame.		269
	Twisted or puckered, affecting wearing comfort.	152	
	More than 1/8 inch above bottom edge of frame.		270
	Width outside of tolerances (see 3.4.3)		271
Labels	Combination identification and size label:		
	Omitted, incorrect, or illegible.	153	
	Misplaced.		273
	Not securely caught in stitching, or stitching through the printing.		274

NOTES:

1/ Material defects are to be classified as indicated only when the condition is one which weakens the fabric or when it is so conspicuously located as to be clearly noticeable. Material defects which are inconspicuous and do not weaken the fabric shall not be classified as defects.

2/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur.

3/ Raw edges not securely caught in the stitching shall be classified as an open seam.

4/ Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.

4.3.2.2 Finished measurements. Any head measurement not in accordance with measurements specified in table III shall be classified as a defect.

4.4 Tests. The inner band shall conform to the requirements specified in paragraph 3.4.1 when tested in accordance with the tests specified in 4.4.1 and 4.4.2.

4.4.1 Stiffness test. The inner band material shall be tested for stiffness at room temperature according to ASTM D2821 - Stiffness, using a span length of 1 inch, a moment weight of 1 inch-pound, and measuring the load at 20 degrees on the angular deflection scale. The specimen shall be 1 inch wide by 4 inches long and it shall be cut from the center part of the band. There shall be three determinations per sample unit.

4.4.2 Bending test. The bending test shall be conducted in accordance with ASTM D6075.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cap frame covered by this specification is for use with the service and dress cap crowns specified in MIL-DTL-18186 and is for the use of personnel of the United States Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. When first article is required and the number of units for the first article (see 3.1 and 6.3).

d. Patterns (see 3.5).

e. Sampling for inspection (see 4.3).

f. Requirements for test reports (see 4.3.1).

g. Packaging requirements (see 5.1).

h. Acceptable quality limits.

6.3 First article. The first article sample should be a preproduction sample. The contracting officer should specify the appropriate type of first article sample and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article sample.

6.4 Subject term (key word) listing.

Embroidery

Enlisted

Officer

Poromeric

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

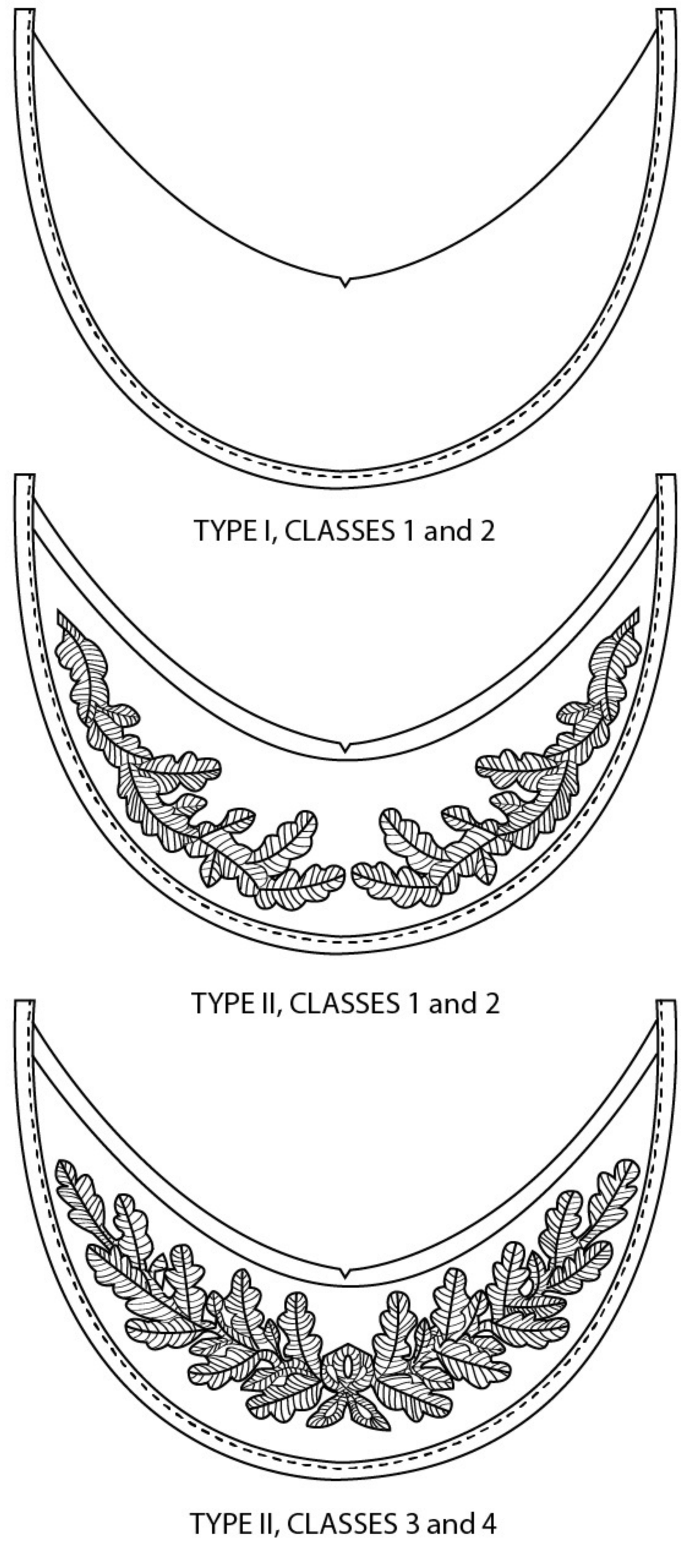
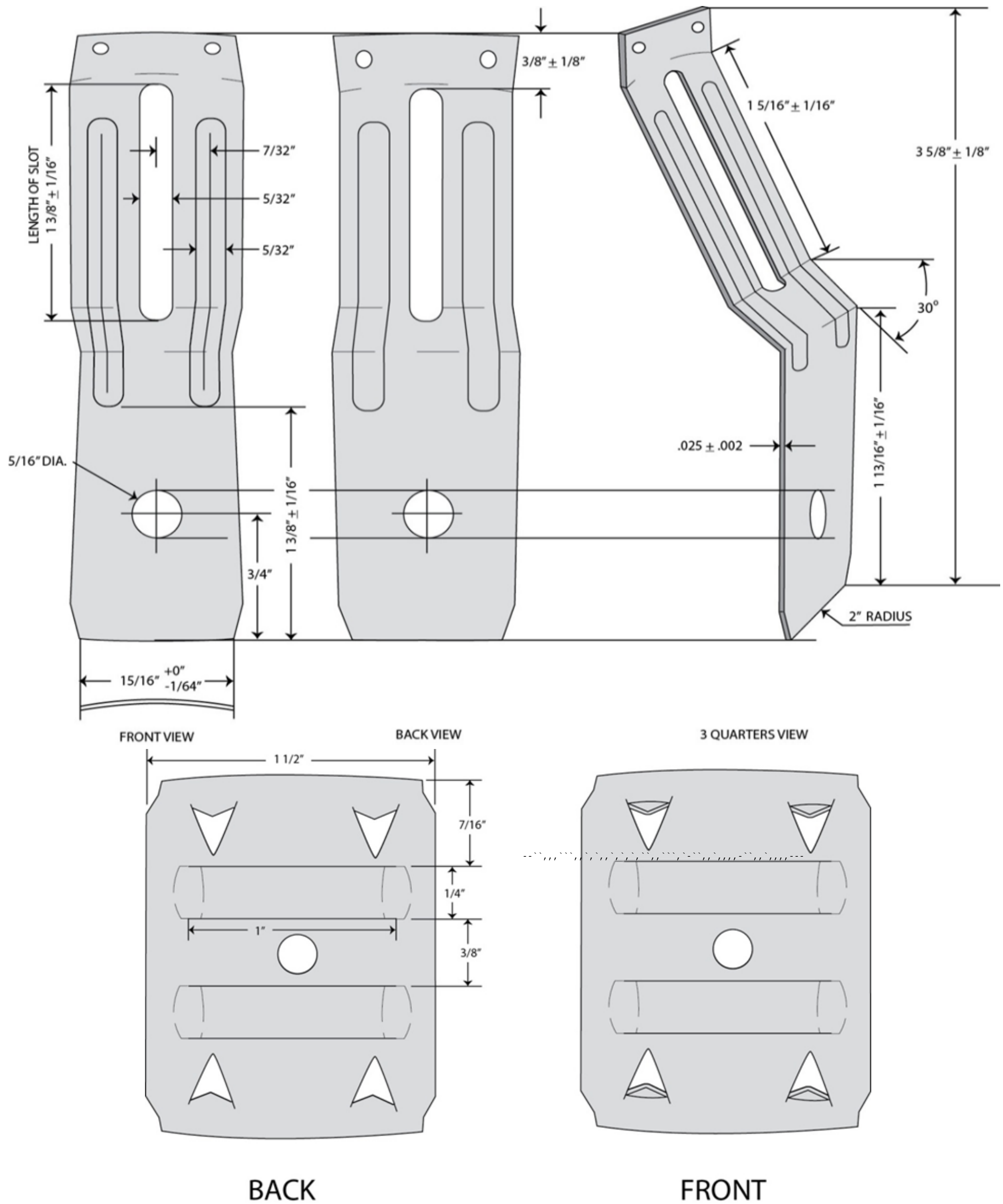
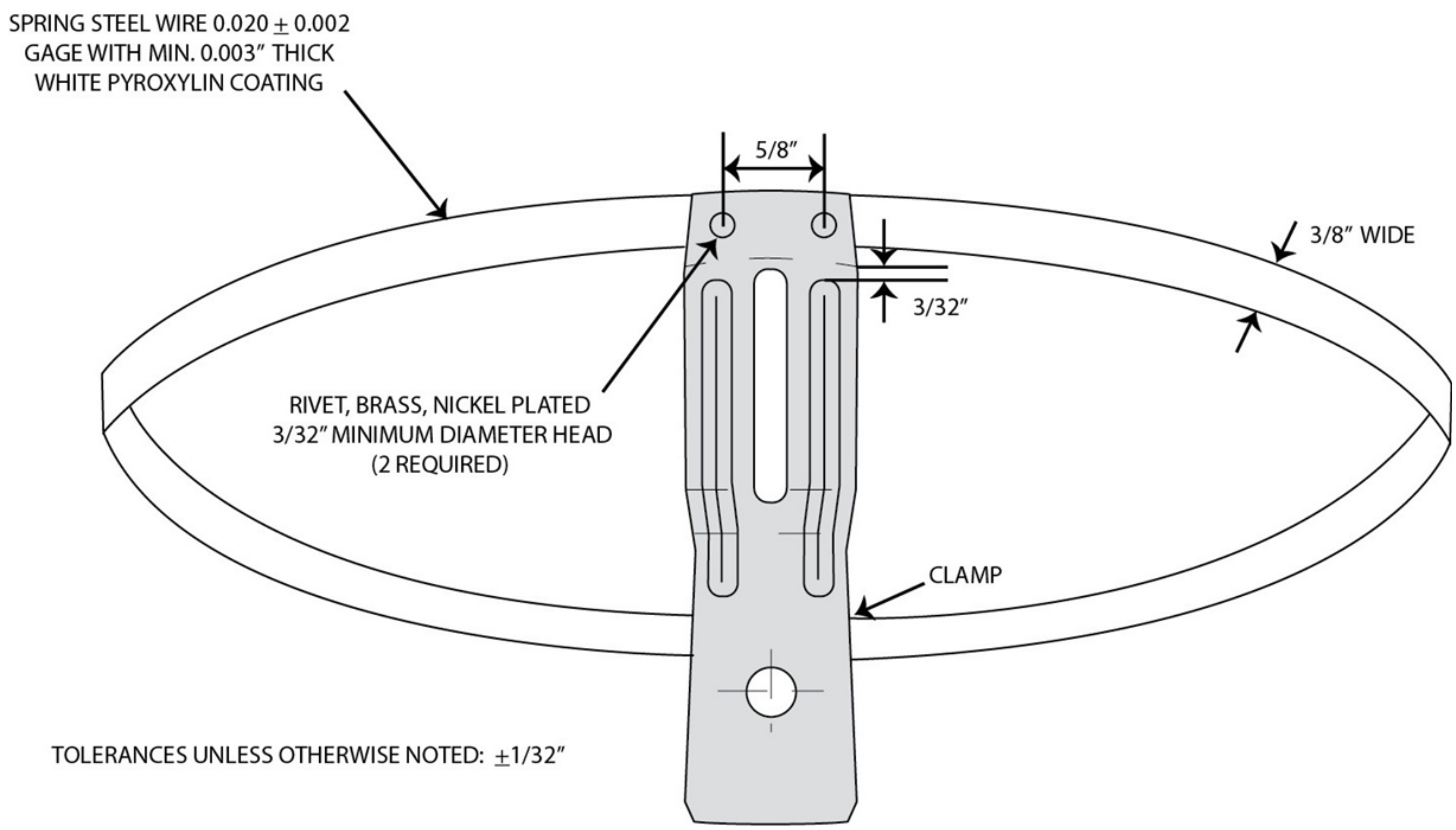


FIGURE 1. Requirements for visors.



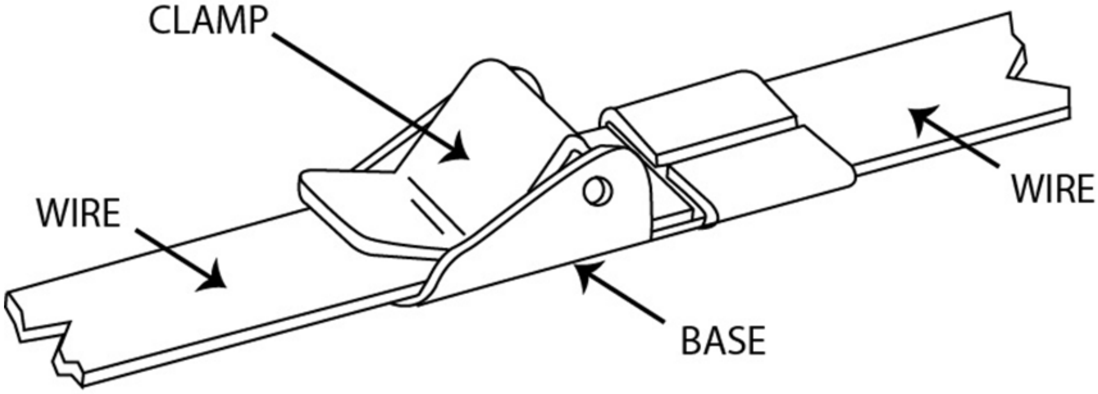
For information only. All measurements in inches.

FIGURE 2. Requirements for support stay and holder.

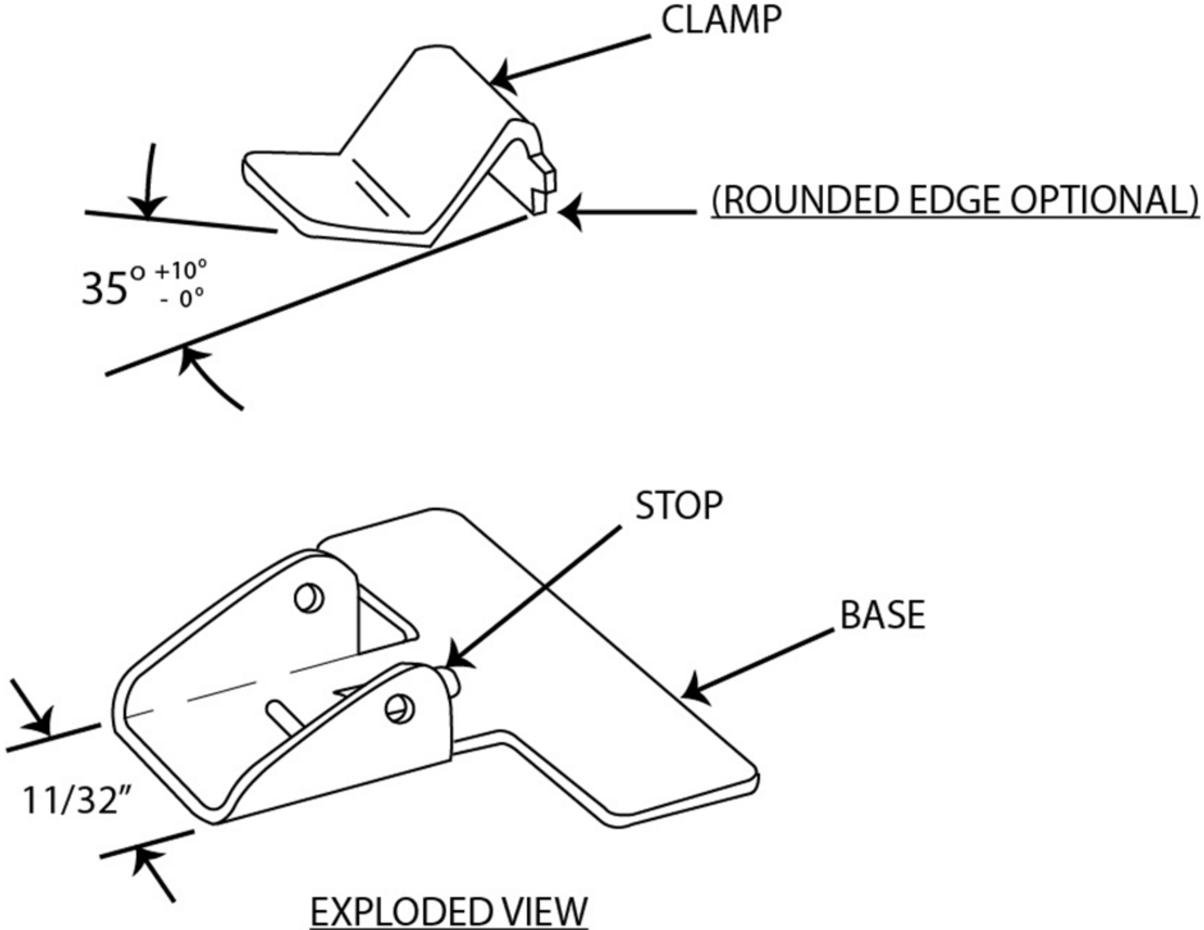


For information only. All measurements are in inches.

FIGURE 3. Requirements for grommet and crown support.



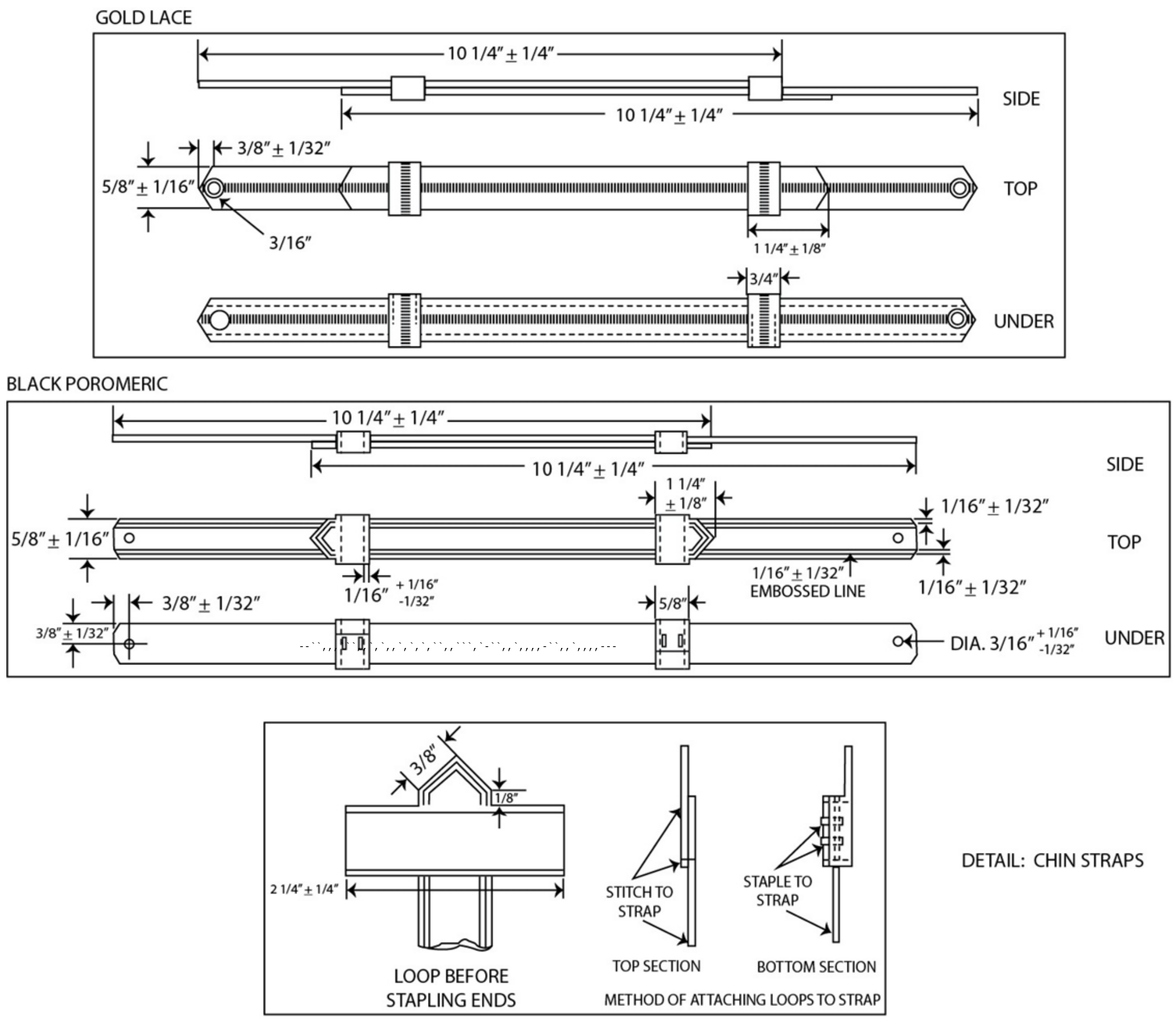
ASSEMBLY



EXPLODED VIEW

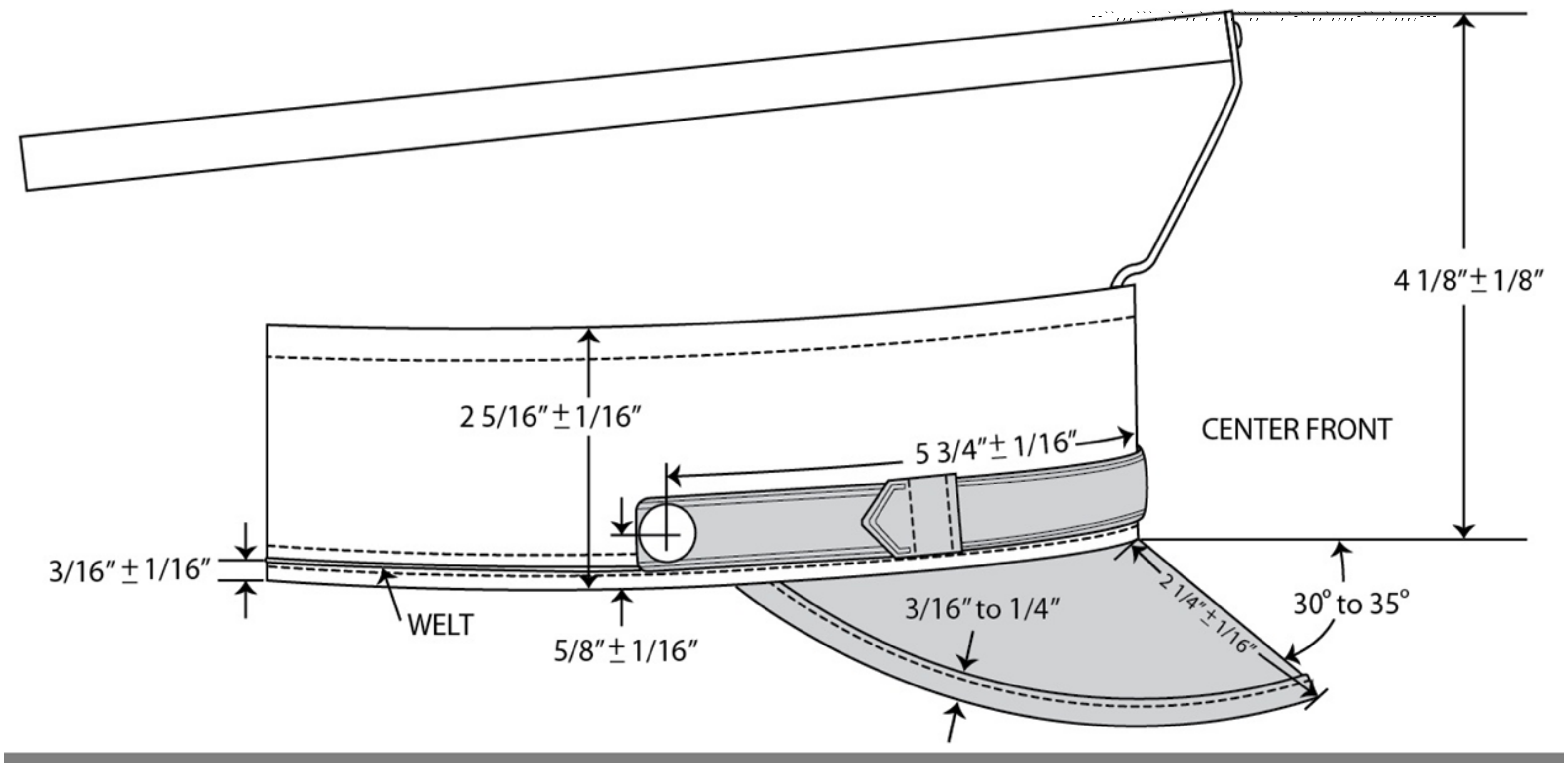
For information only. All measurements are in inches.

FIGURE 4. Exploded view – requirements for clamp and assembly.



For information only. All measurements are in inches.

FIGURE 5. Requirements for chin straps.



For information only. All measurements are in inches.

FIGURE 6. Requirements for frame – side view.

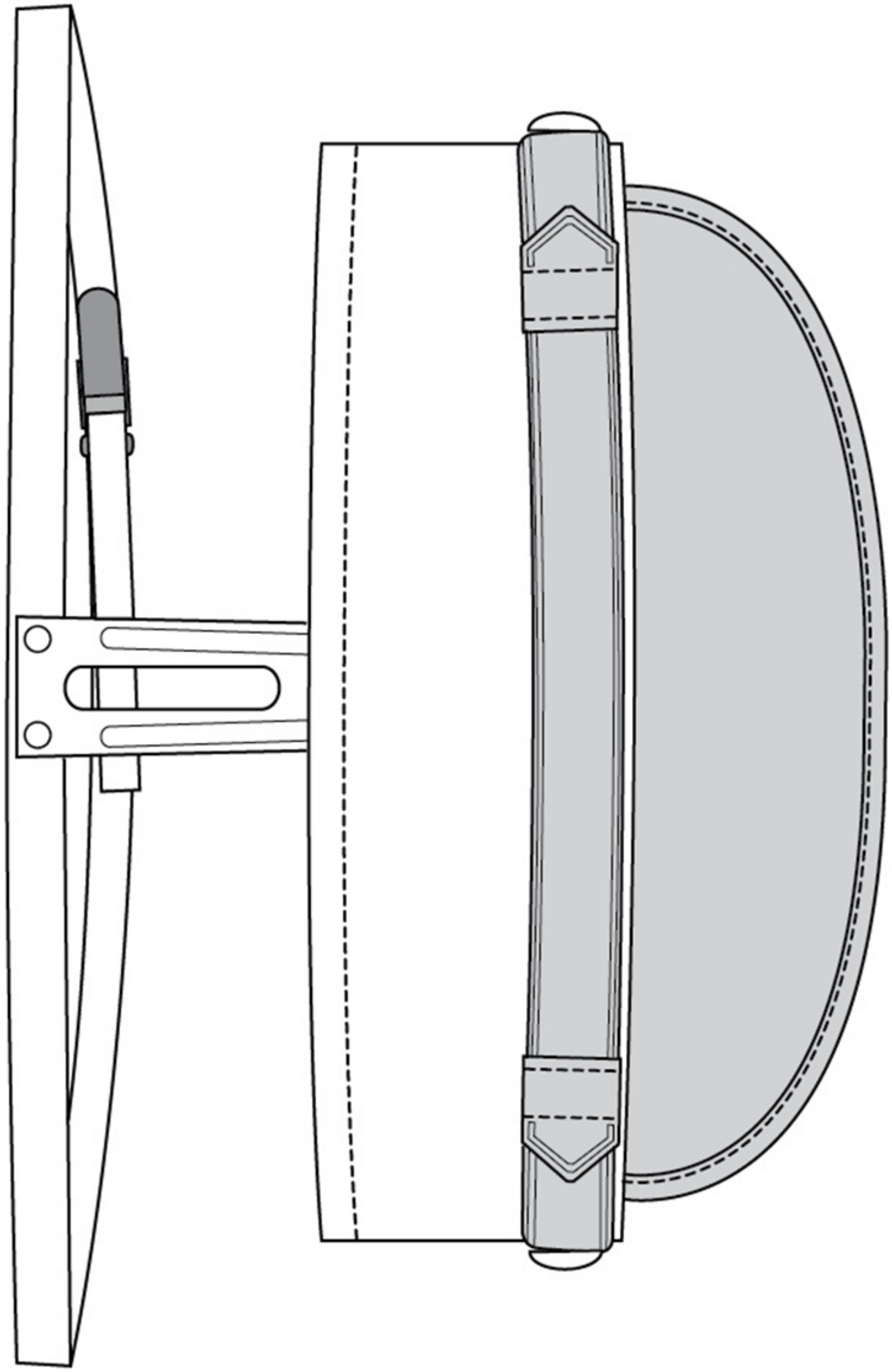


FIGURE 7. Frame – front view.

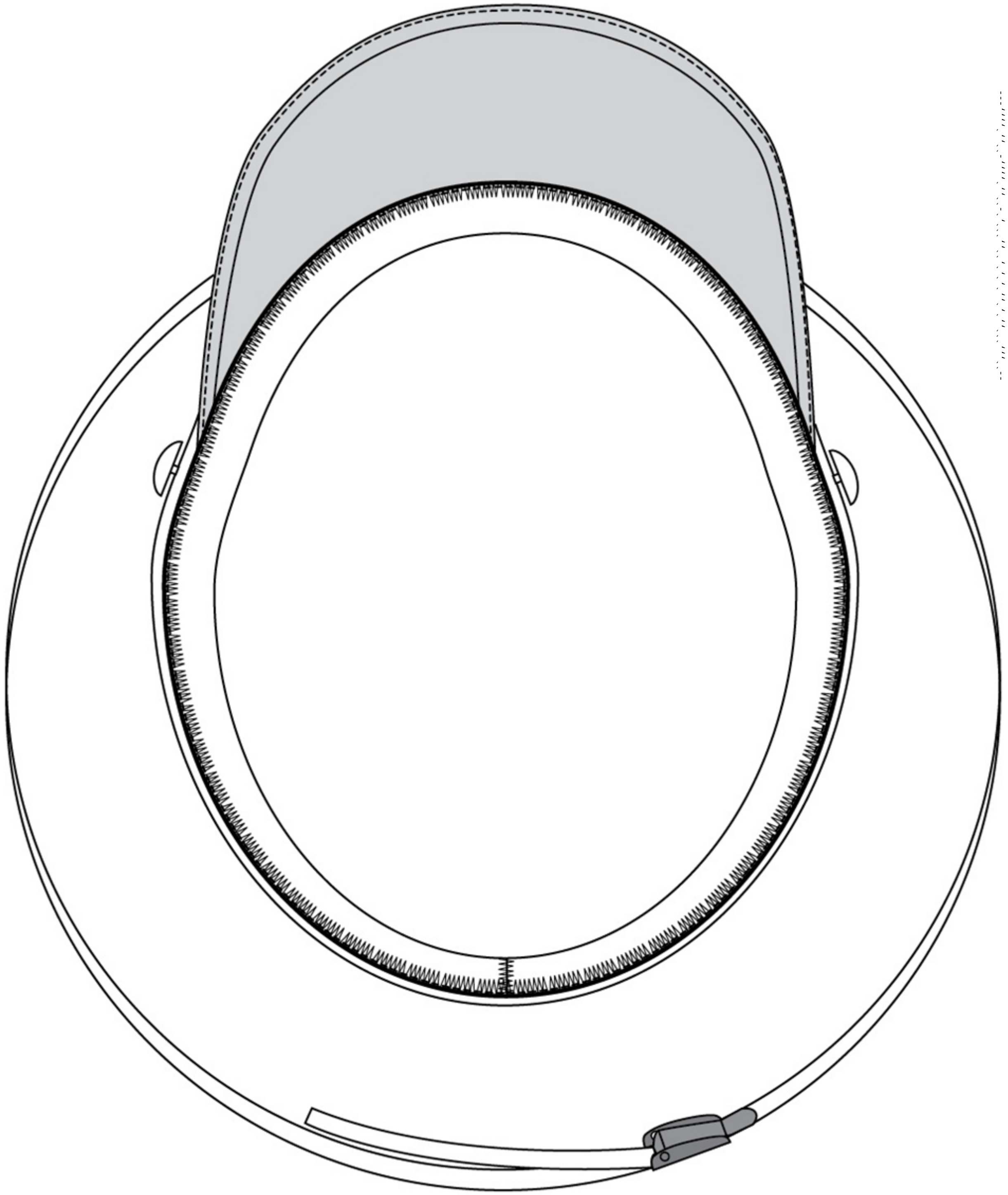


FIGURE 8. Frame – inside band view.

CONCLUDING MATERIAL

Custodians:
Navy – MC
DLA – CT

Preparing activity:
Navy – MC
(Project 8405-2025-001)

Review activity:
Navy – NU

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.